Wireshark Filter

1. Wireshark Filter

- 캡처 필터(Capture Filter)

  - 패킷이 캡처 될 때 지정

  - 지정된 표현식에 포함/제외된 패킷만 캡처

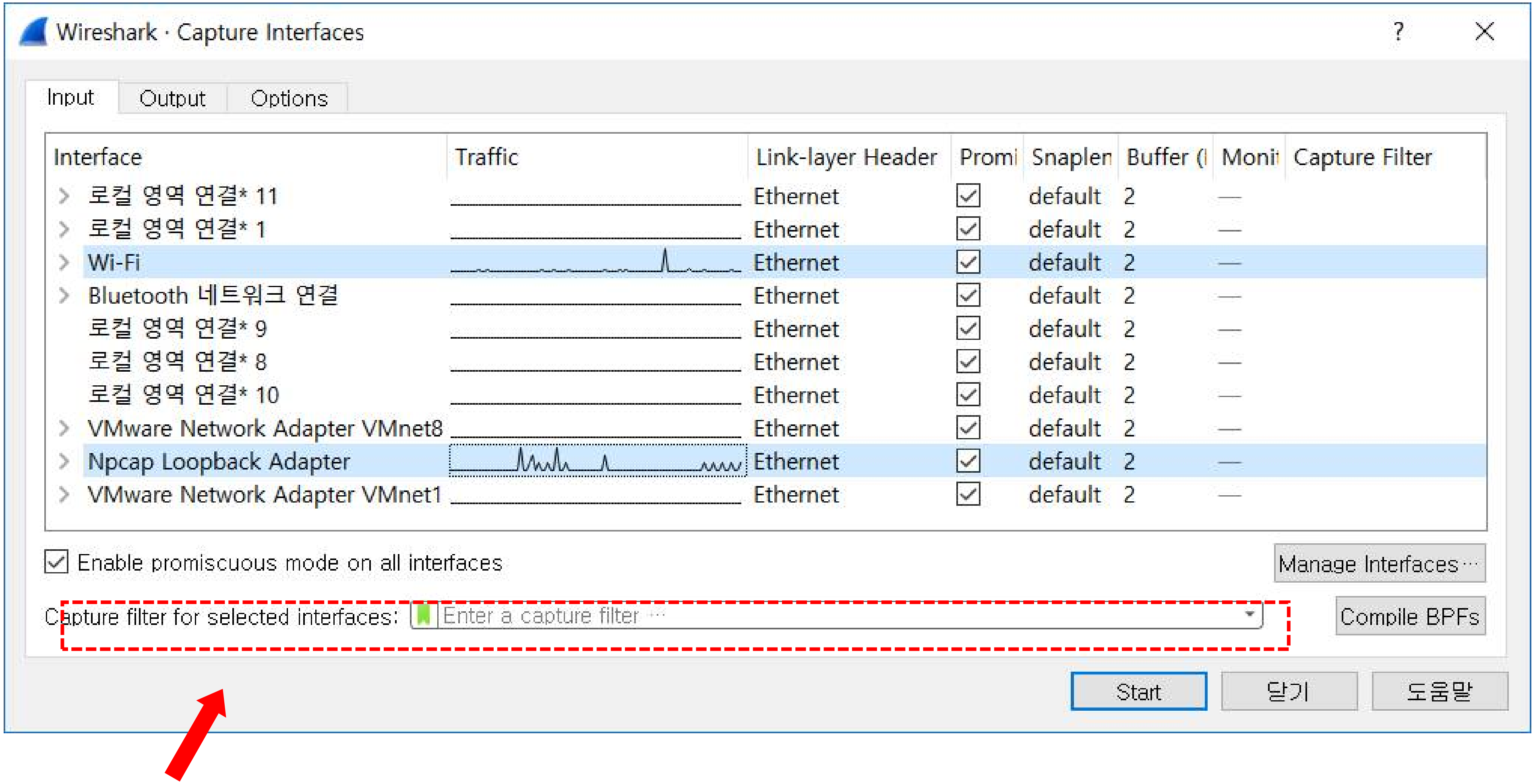
- 디스플레이 필터(Display Filter)

  - 원하지 않는 패킷을 숨김

  - 지정된 표현식을 기반으로 원하는 패킷을 보기

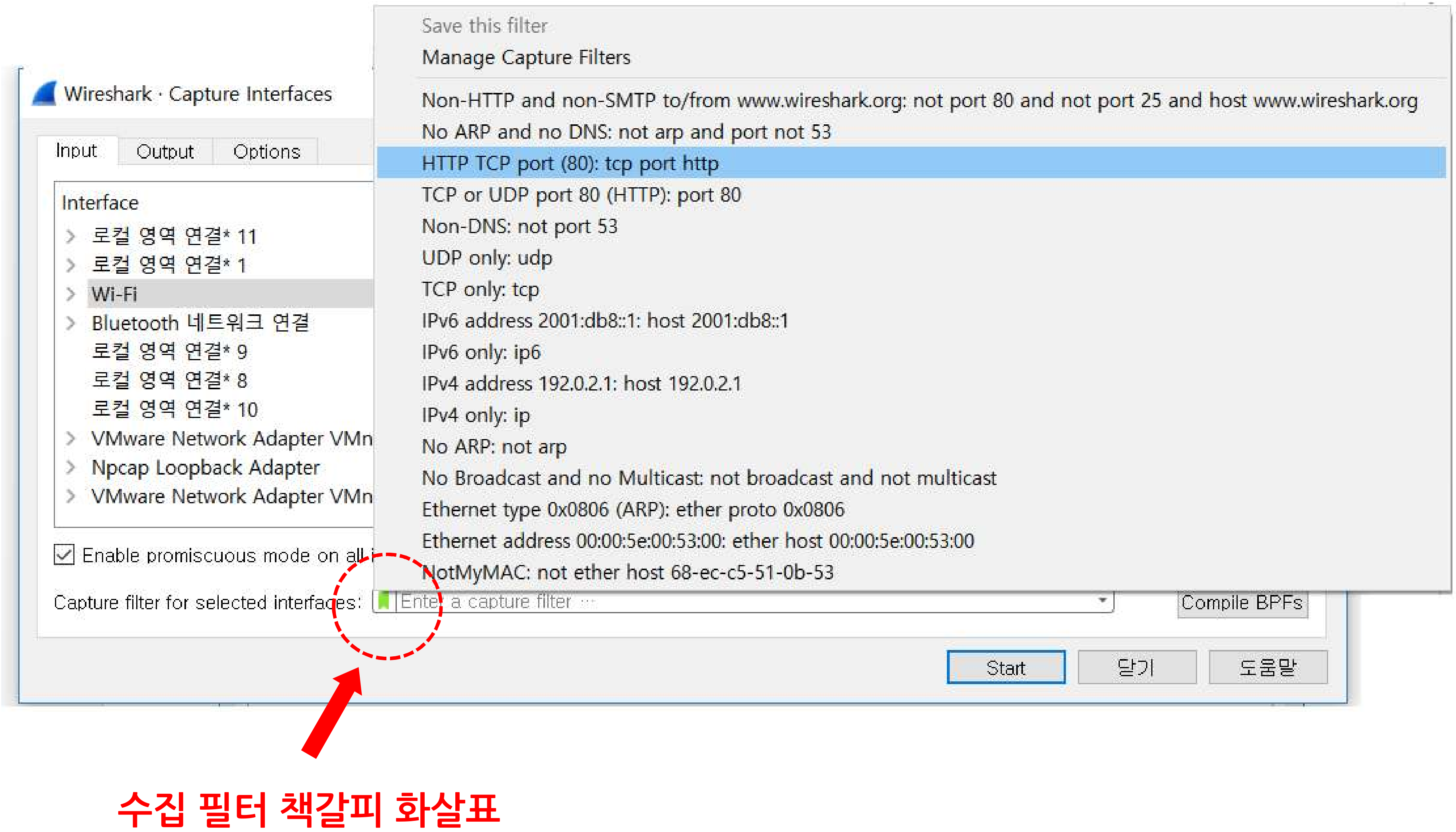
1) Capture Filter

- Capture Option 창에서 수집 필터 적용



Capture Option 창에서 수집 필터 적용

❶ Capture Option > 수집 필터 책갈피 화살표



주소 기반의 트래픽 수집

❶ 특정 IP 주소에서/로 오는 트래픽 수집

- host 10.3.1.1
- host 2406:da00:ff00::6b16:f02d
- not host 10.3.1.1
- src host 10.3.1.1
- dst host 10.3.1.1
- host 10.3.1.1 or host 10.3.1.2
- host www.espn.com

주소 기반의 트래픽 수집

❷ IP 주소 범위에서/로 오는 트래픽 수집

- net 10.3.0.0/16
- net 10.3.0.0 mask 255.255.0.0
- ipv6 net 2406:da00:ff00::/64
- not dst net 10.3.0.0/16
- dst net 10.3.0.0/16
- src net 10.3.0.0/16

주소 기반의 트래픽 수집

❸ 브로드캐스트 또는 멀티캐스트 트래픽 수집

- ip broadcast
- ip multicast
- dst host ff02::1
- dst host ff02::2

주소 기반의 트래픽 수집

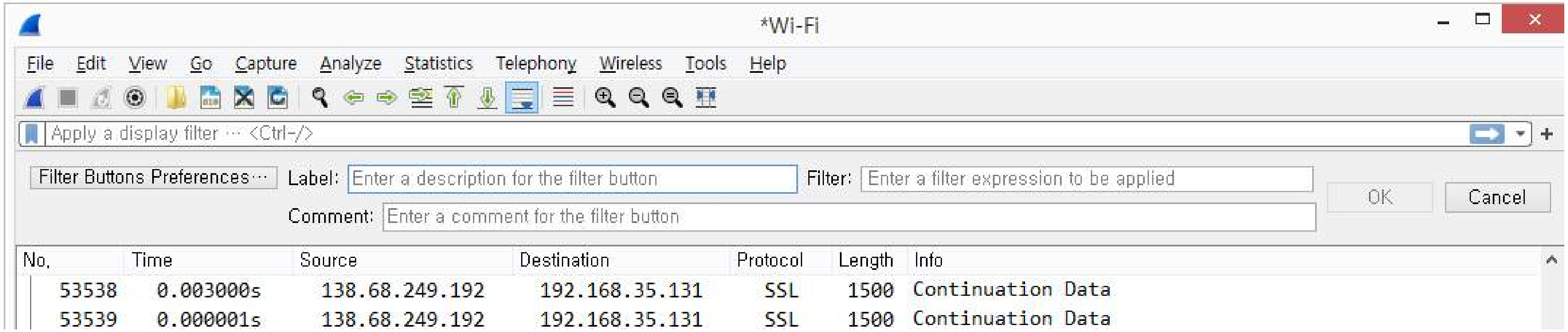
❹ MAC 주소 기반의 트래픽 수집

- ether host 00:08:15:00:08:15
- ether src 00:08:15:00:08:15
- ether dst 00:08:15:00:08:15
- not ether host 00:08:15:00:08:15

특정 애플리케이션에 대한 트래픽 수집

- port 53
- not port 53
- port 80
- udp port 67
- tcp port 21
- portrange 1-80
- tcp portrange 1-80
- port 20 or port 21
- host 10.3.1.1 and port 80
- host 10.3.1.1 and not port 80
- udp src port 68 and udp dst port 67

2) 디스플레이 필터


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter … <Ctrl-/>
Filter Buttons Preferences…
Label: Enter a description for the filter button
Filter: Enter a filter expression to be applied
OK
Cancel
Comment: Enter a comment for the filter button
No. Time Source Destination Protocol Length Info
53538 0.003000s 138.68.249.192 192.168.35.131 SSL 1500 Continuation Data
53539 0.000001s 138.68.249.192 192.168.35.131 SSL 1500 Continuation Data

적절한 디스플레이 필터 문법 사용

- 간단한 디스플레이 필터 문법

  arp

  ip

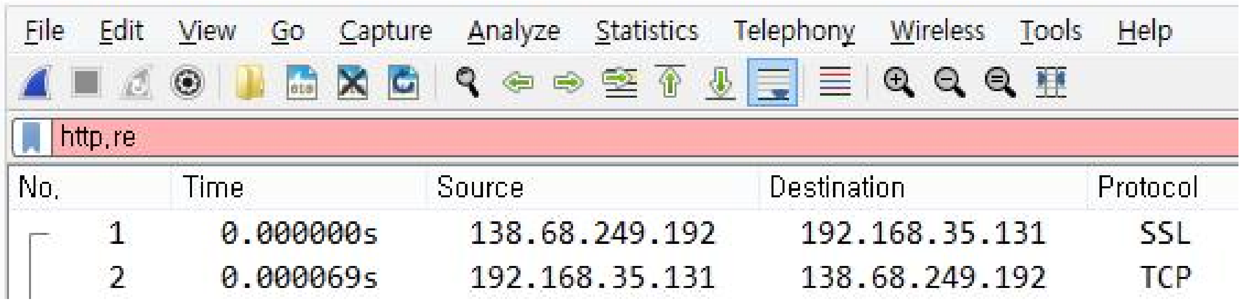
  ipv6

  tcp

적절한 디스플레이 필터 문법 사용

<<디스플레이 필터 오류 탐지 메커니즘 >>

- 대소문자 구분
- 적색 배경
  - 문법 검사 실패
  - 동작하지 않음



- 녹색 배경
  - 문법 이상 없음
  - '논리 검사'는 하지 않음 (예) http && udp
- 황색 배경
  - 필터가 원하는 대로 동작하지 않는 것을 경고 (예) ip.addr != 10.1.1.1

디스플레이필터와 연산자 비교

연산자	영어표기	예제
==	eq	ip.src == 10.2.2.2
!=	ne	tcp.srcport != 80
>	gt	frame.time_relative > 1
<	lt	tcp.window_size < 1460
>=	ge	dns.count.answers >=10
<=	lt	ip.ttl < 10
	contains	http contain “GET”

캡처 필터 vs 디스플레이 필터

캡처 필터 구문 예제	디스플레이 필터 예제
host 172.16.1.1	ip.host == 172.16.1.1
src host 172.16.1.1	ip.src ==172.16.1.1
dst host 172.16.1.1	ip.dst ==172.16.1.1
port 8080	tcp.port == 8080
!port 8080	!tcp.port = 8080

❶ 단순 IP 주소 호스트에게/부터의 트래픽 필터링

- ip.addr == 10.3.1.1
- !ip.addr ==10.3.1.1
- ipv6.addr == 2406:da00:ff00::6b16:f02d
- ip.src==10.3.1.1
- ip.dst == 10.3.1.1
- ip.host == www.wireshark.org

❷ 주소 범위에게/부터의 트래픽 필터링

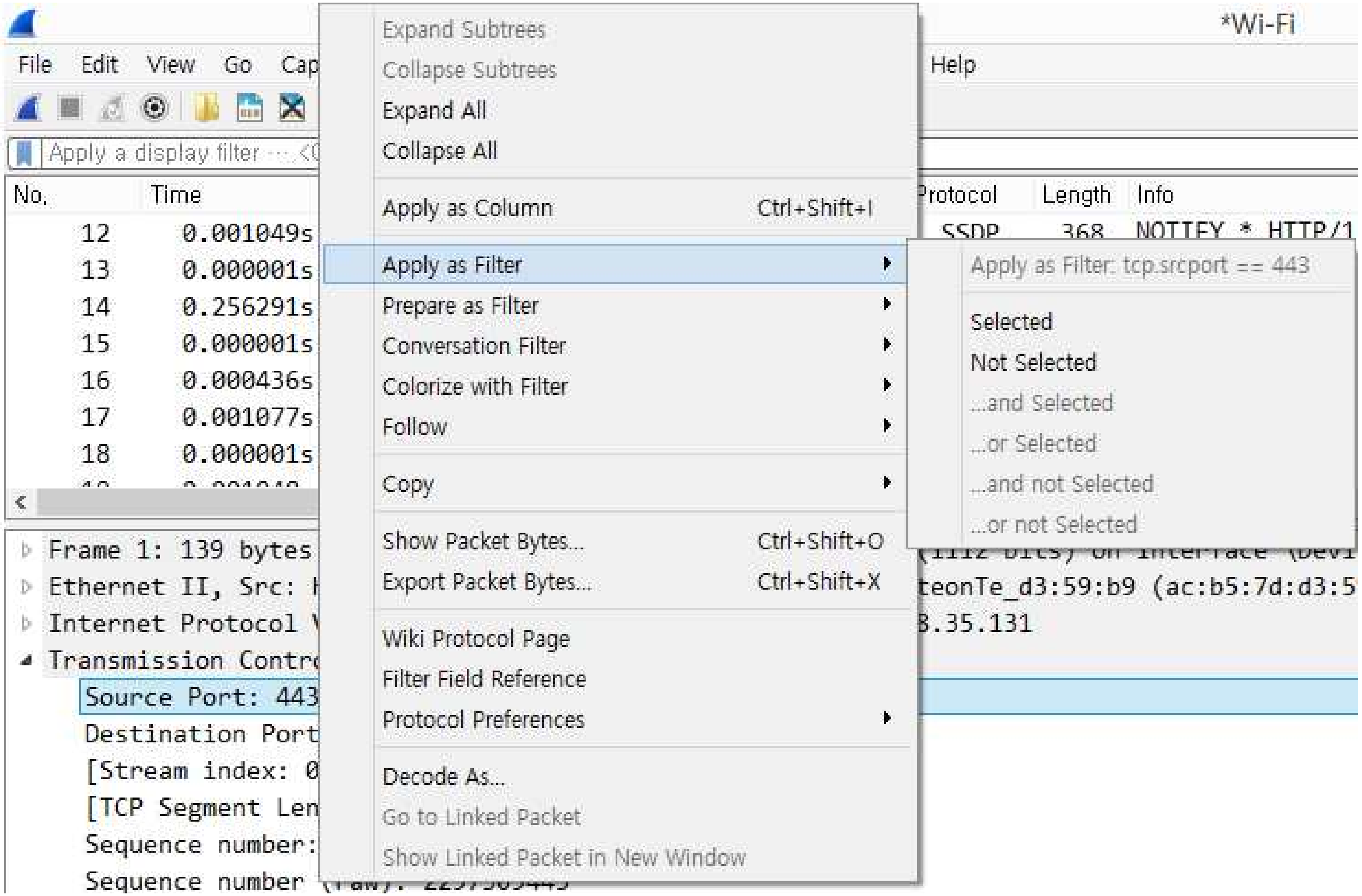
- ip.addr > 10.3.0.1 && ip.addr < 10.3.0.5
- (ip.addr >= 10.3.0.1 && ip.addr <= 10.3.0.6) && !ip.addr ==10.3.0.3
- ipv6.addr == fe80:: && ipv6.addr < fec0::

❸ IP 서브넷에서/으로부터 트래픽 필터링

- ip.addr == 10.3.0.0/16
- ip.addr == 10.3.0.0/16 && !ip.addr ==10.3.0.3
- !ip.addr == 10.3.0.0/16 && !ip.addr ==10.2.0.0/16

- Apply as Filter

- Prepare as Filter



Selected

A == B

Not Selected

!(A==B)

… and Selected

(A==B) && (C…)

... or Selected

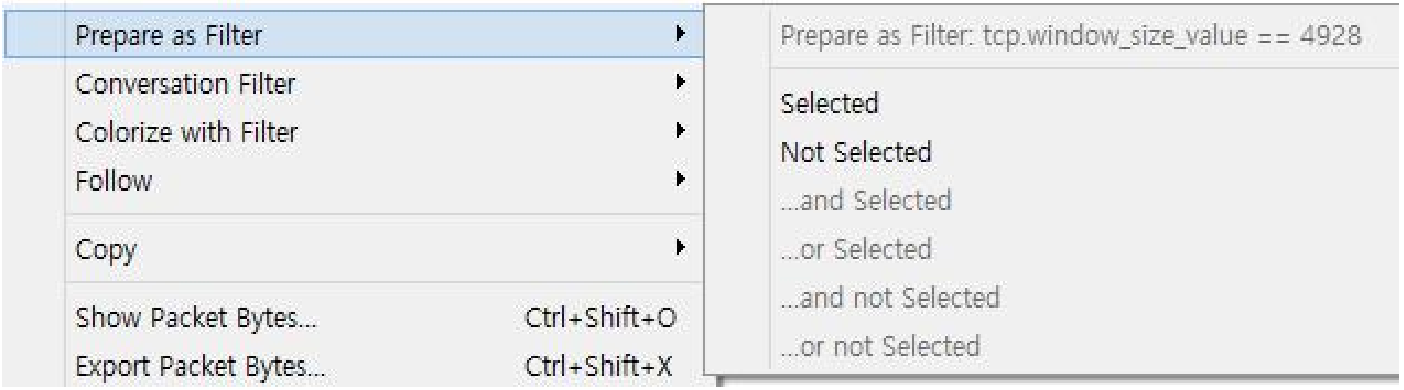
(A==B) || (C…)

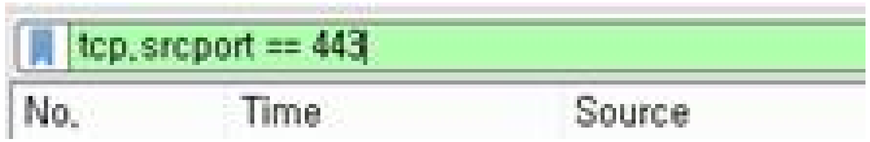
… and Not Selected

(A==B) && !(C…)

… or Not Selected

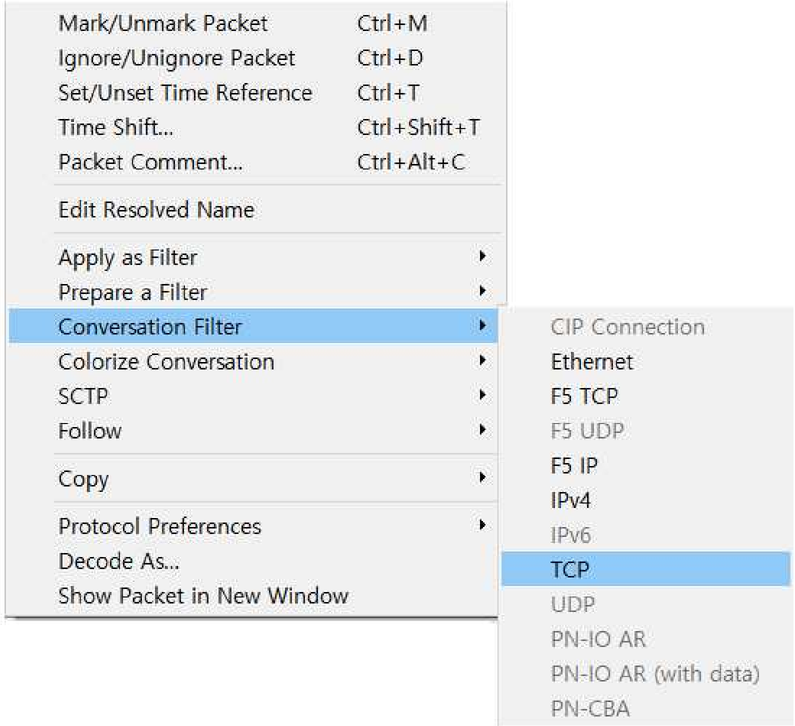
(A==B) || !(C…)



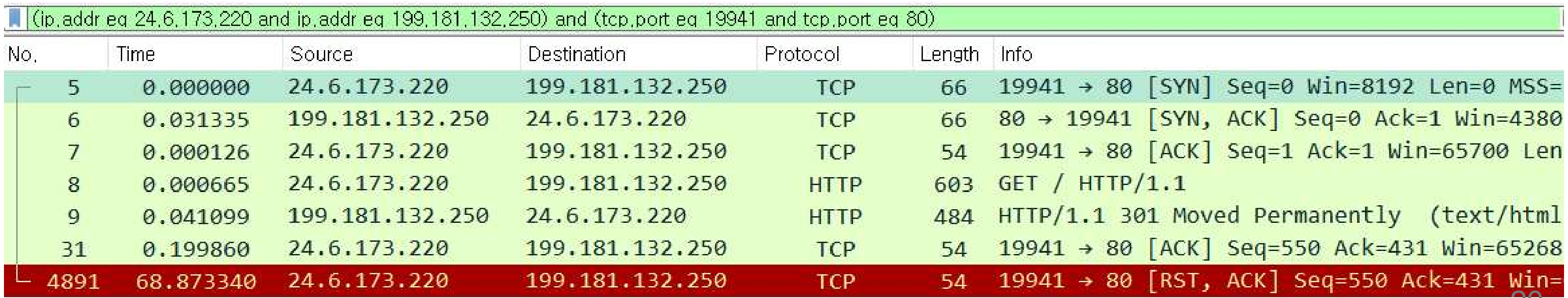


3) Conversation Filter

❶ 패킷 리스트 > 패킷선택 > 오른쪽 마우스 클릭 > Conversation Filter > TCP



추적파일 : http-espn101.pcapng

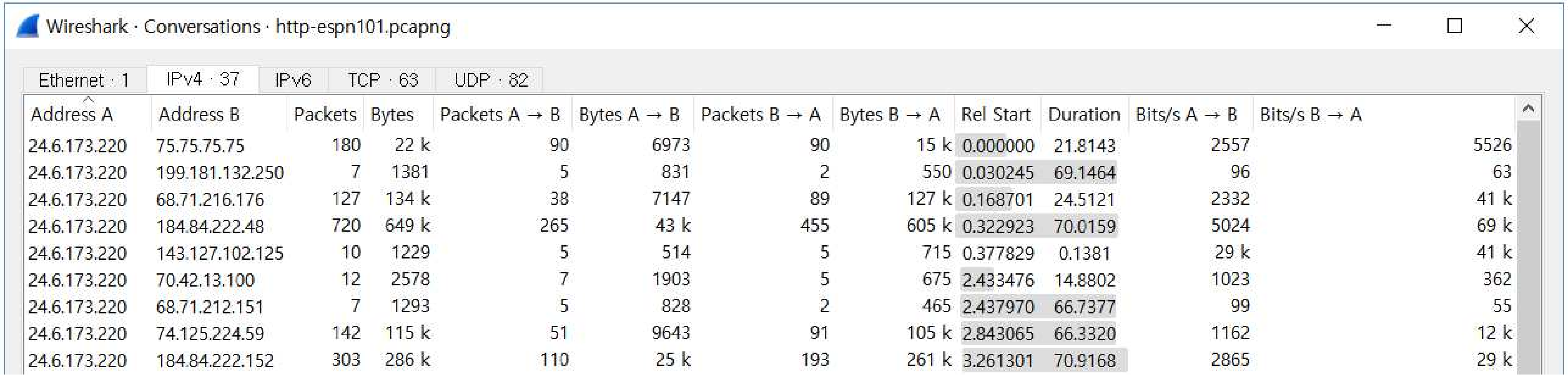


(ip.addr eq 24.6.173.220 and ip.addr eq 199.181.132.250) and (tcp.port eq 19941 and tcp.port eq 80)

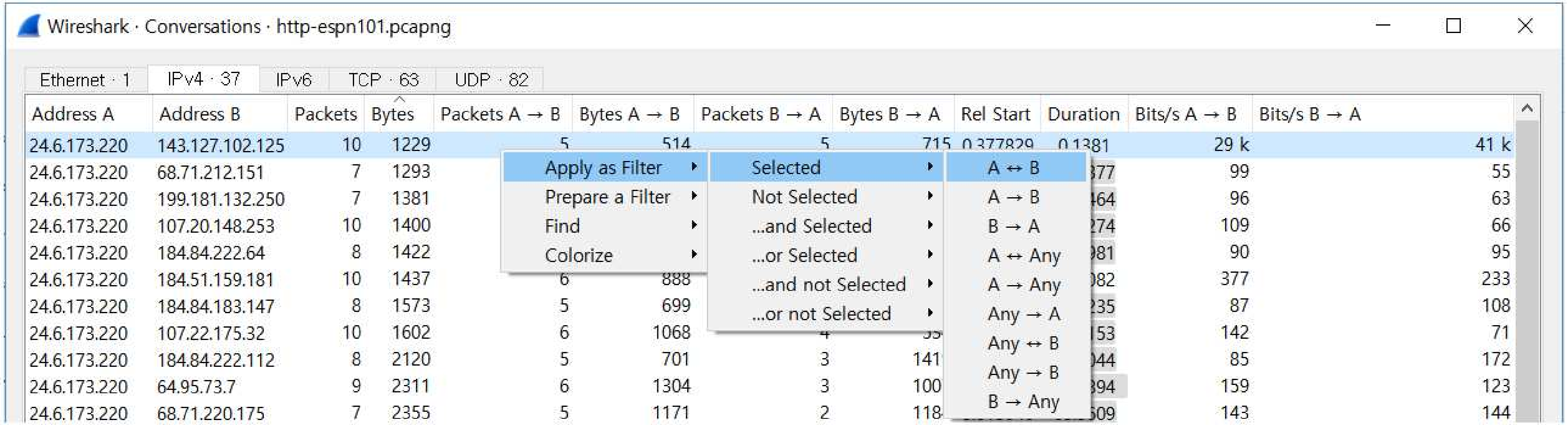
No.	Time	Source	Destination	Protocol	Length	Info
5	0.000000	24.6.173.220	199.181.132.250	TCP	66	19941 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=
6	0.031335	199.181.132.250	24.6.173.220	TCP	66	80 → 19941 [SYN, ACK] Seq=0 Ack=1 Win=4380
7	0.000126	24.6.173.220	199.181.132.250	TCP	54	19941 → 80 [ACK] Seq=1 Ack=1 Win=65700 Len
8	0.000665	24.6.173.220	199.181.132.250	HTTP	603	GET / HTTP/1.1
9	0.041099	199.181.132.250	24.6.173.220	HTTP	484	HTTP/1.1 301 Moved Permanently (text/html
31	0.199860	24.6.173.220	199.181.132.250	TCP	54	19941 → 80 [ACK] Seq=550 Ack=431 Win=65268
4891	68.873340	24.6.173.220	199.181.132.250	TCP	54	19941 → 80 [RST, ACK] Seq=550 Ack=431 Win=

❷ Statistics > Conversation Filter

추적파일 : http-espn101.pcapng



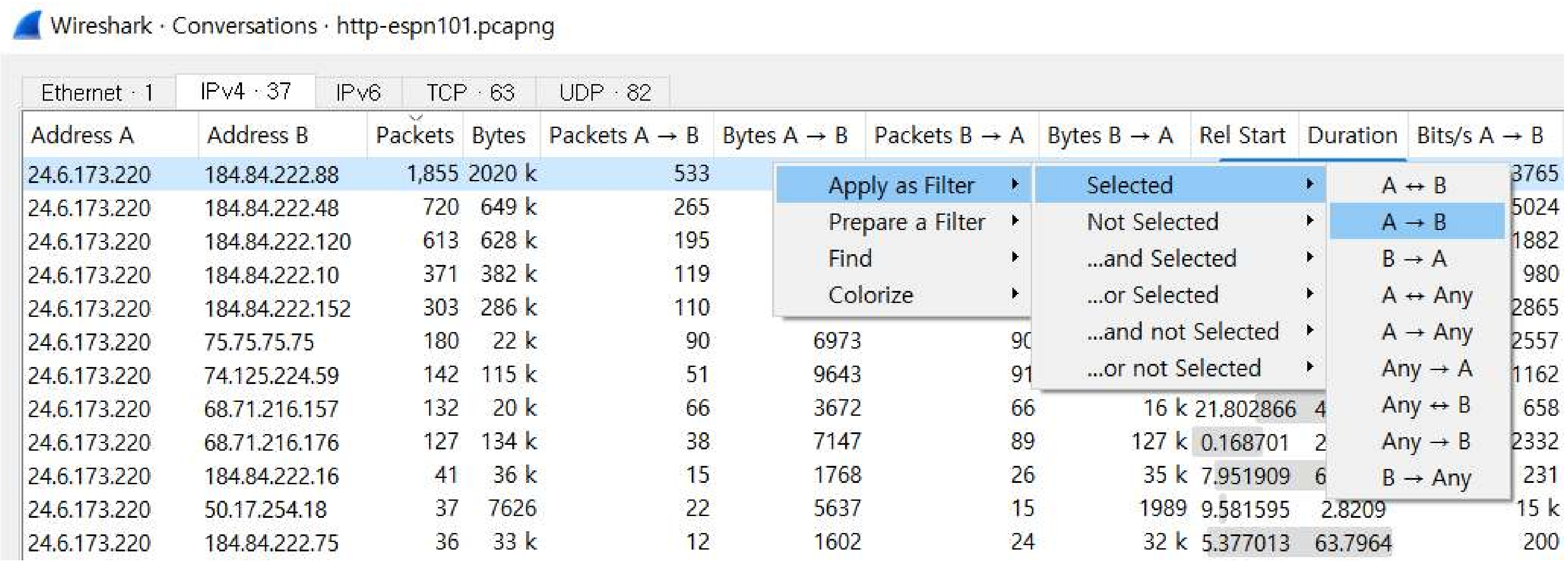
Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
24.6.173.220	75.75.75.75	180	22 k	90	6973	90	15 k	0.000000	21.8143	2557	5526
24.6.173.220	199.181.132.250	7	1381	5	831	2	550	0.030245	69.1464	96	63
24.6.173.220	68.71.216.176	127	134 k	38	7147	89	127 k	0.168701	24.5121	2332	41 k
24.6.173.220	184.84.222.48	720	649 k	265	43 k	455	605 k	0.322923	70.0159	5024	69 k
24.6.173.220	143.127.102.125	10	1229	5	514	5	715	0.377829	0.1381	29 k	41 k
24.6.173.220	70.42.13.100	12	2578	7	1903	5	675	2.433476	14.8802	1023	362
24.6.173.220	68.71.212.151	7	1293	5	828	2	465	2.437970	66.7377	99	55
24.6.173.220	74.125.224.59	142	115 k	51	9643	91	105 k	2.843065	66.3320	1162	12 k
24.6.173.220	184.84.222.152	303	286 k	110	25 k	193	261 k	3.261301	70.9168	2865	29 k



Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
24.6.173.220	143.127.102.125	10	1229	5	514	5	715	0.377829	0.1381	29 k	41 k
24.6.173.220	68.71.212.151	7	1293						377	99	55
24.6.173.220	199.181.132.250	7	1381						464	96	63
24.6.173.220	107.20.148.253	10	1400						274	109	66
24.6.173.220	184.84.222.64	8	1422						981	90	95
24.6.173.220	184.51.159.181	10	1437	6	888				082	377	233
24.6.173.220	184.84.183.147	8	1573	5	699				235	87	108
24.6.173.220	107.22.175.32	10	1602	6	1068	4			153	142	71
24.6.173.220	184.84.222.112	8	2120	5	701	3	141		044	85	172
24.6.173.220	64.95.73.7	9	2311	6	1304	3	100		394	159	123
24.6.173.220	68.71.220.175	7	2355	5	1171	2	118		509	143	144

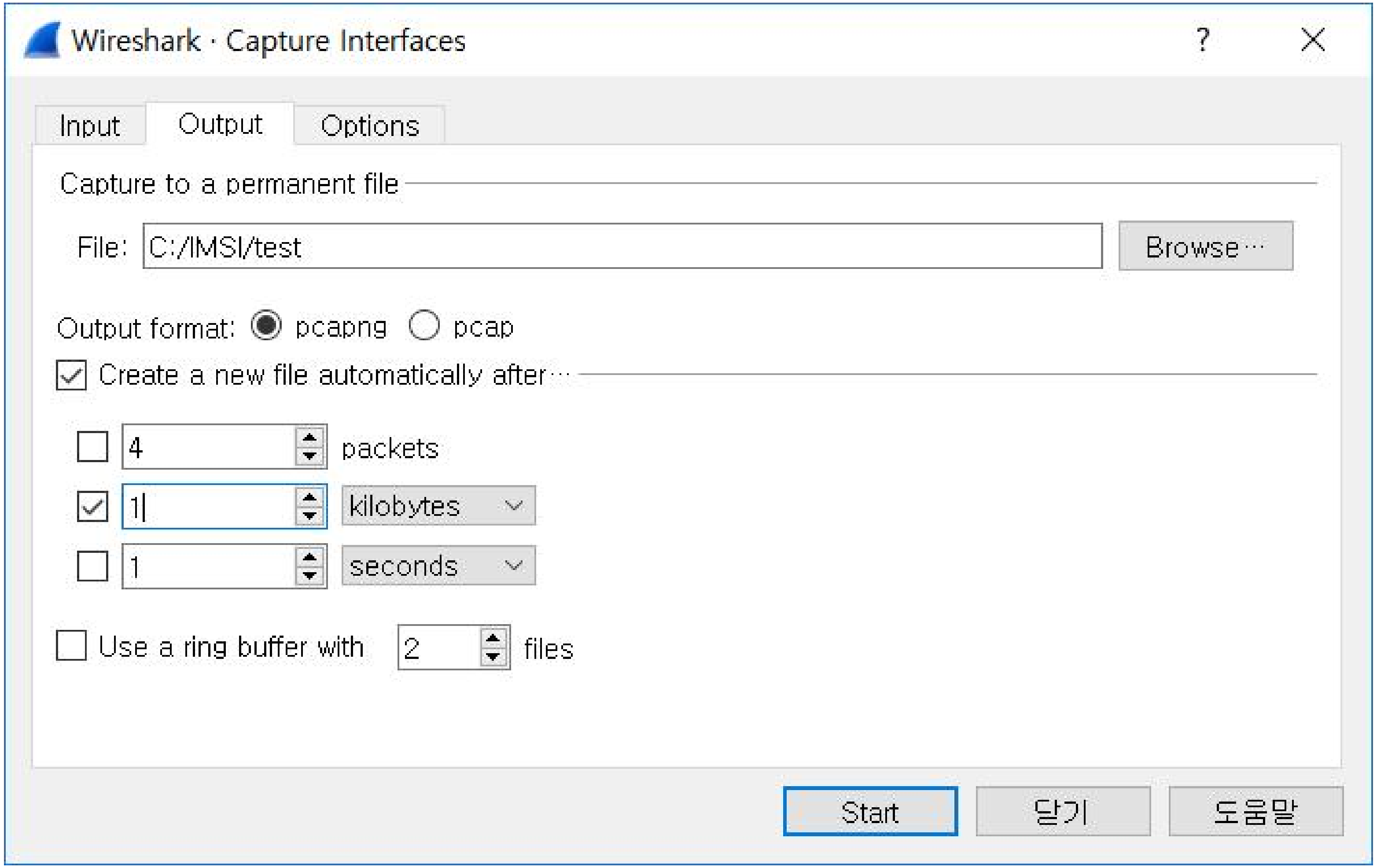
❷ Statistics > Conversation Filter

- Packets 필드를 기준으로 내림 차순으로 정렬
- 첫 번째 패킷 선택
- 오른쪽 마우스 클릭 > Apply as Filter > Selected > A→ B



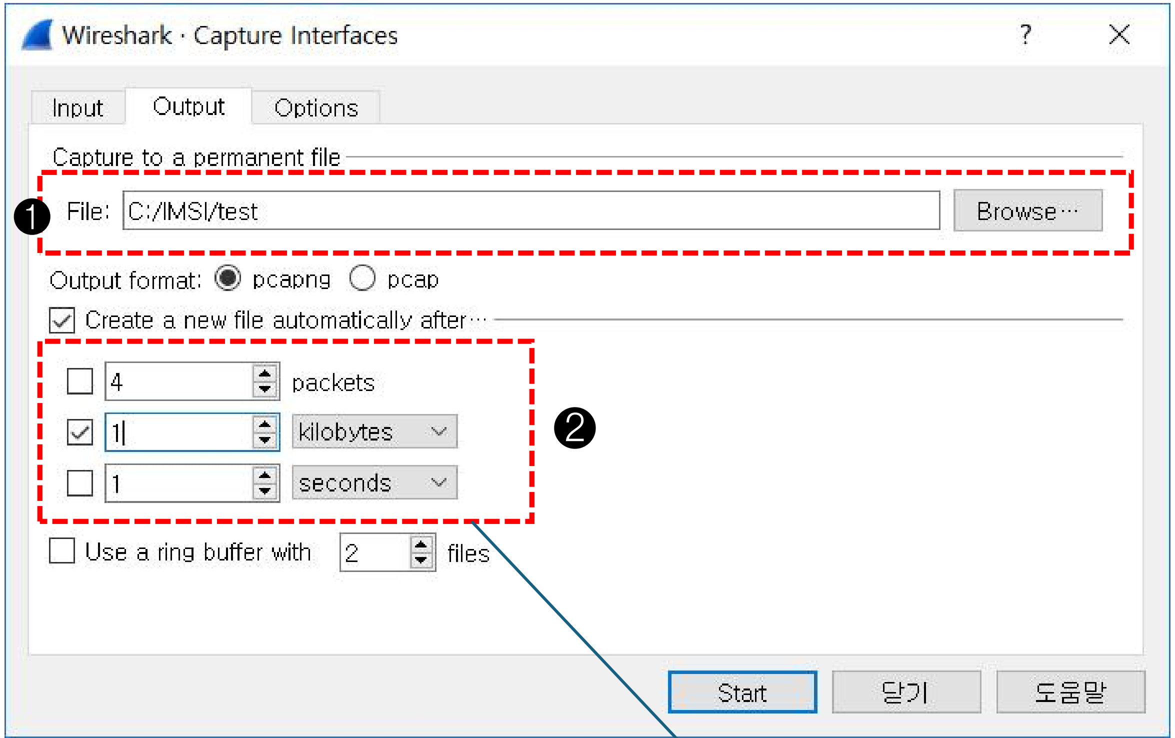
파일 집합으로 수집

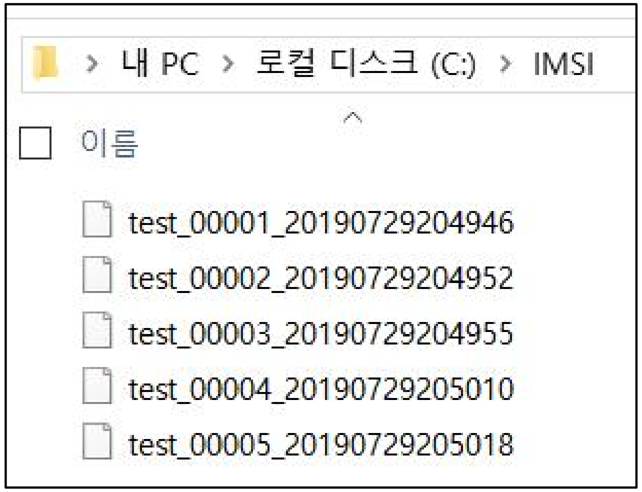
- Capture Options > Output Tab > Create a new file automatically after…



파일 집합으로 수집

- Capture Options > Output Tab > Create a new file automatically after…





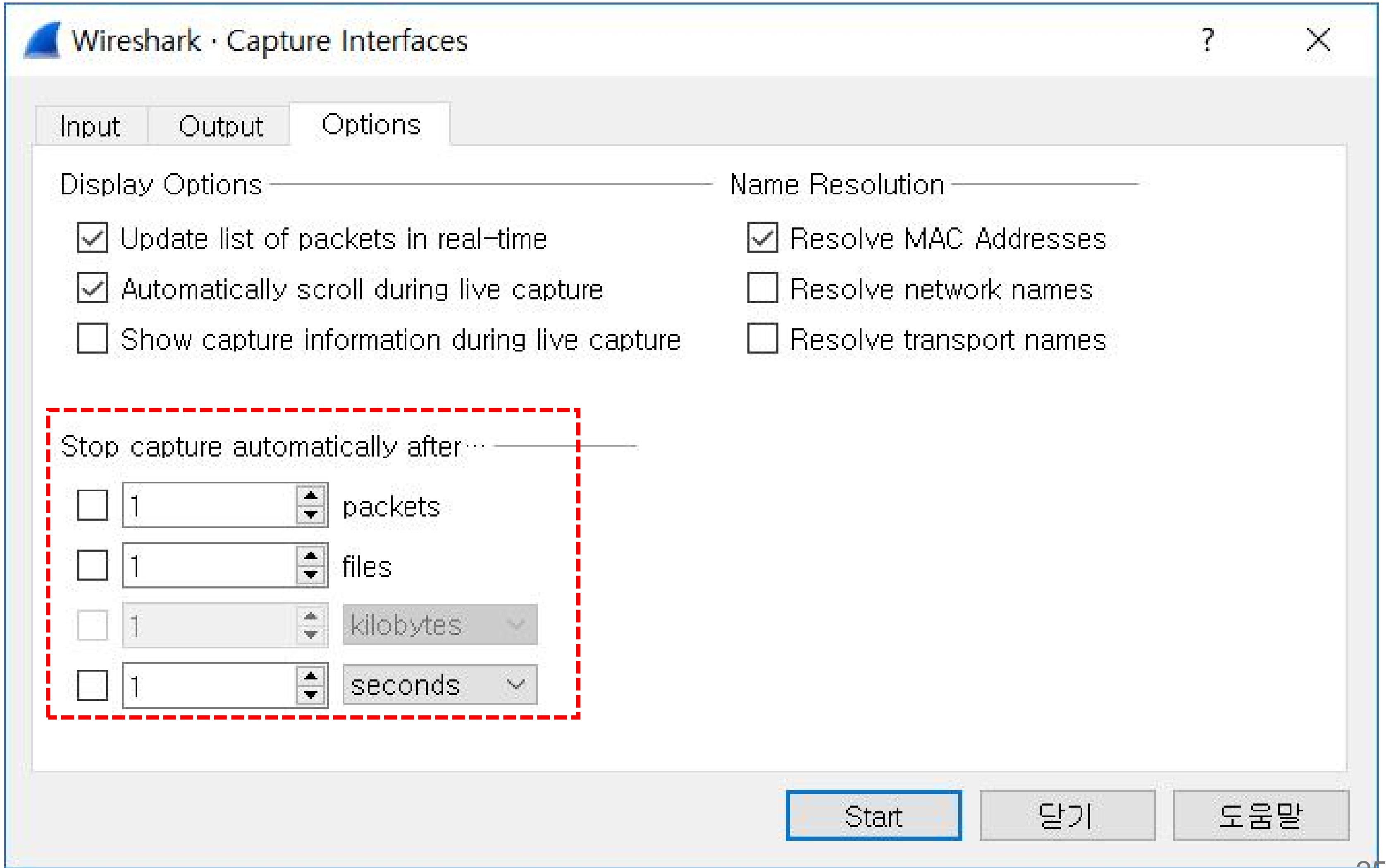
- File > Open
  - File > File set > List Files

파일당 4개의 패킷
1MB 파일 크기
1초마다

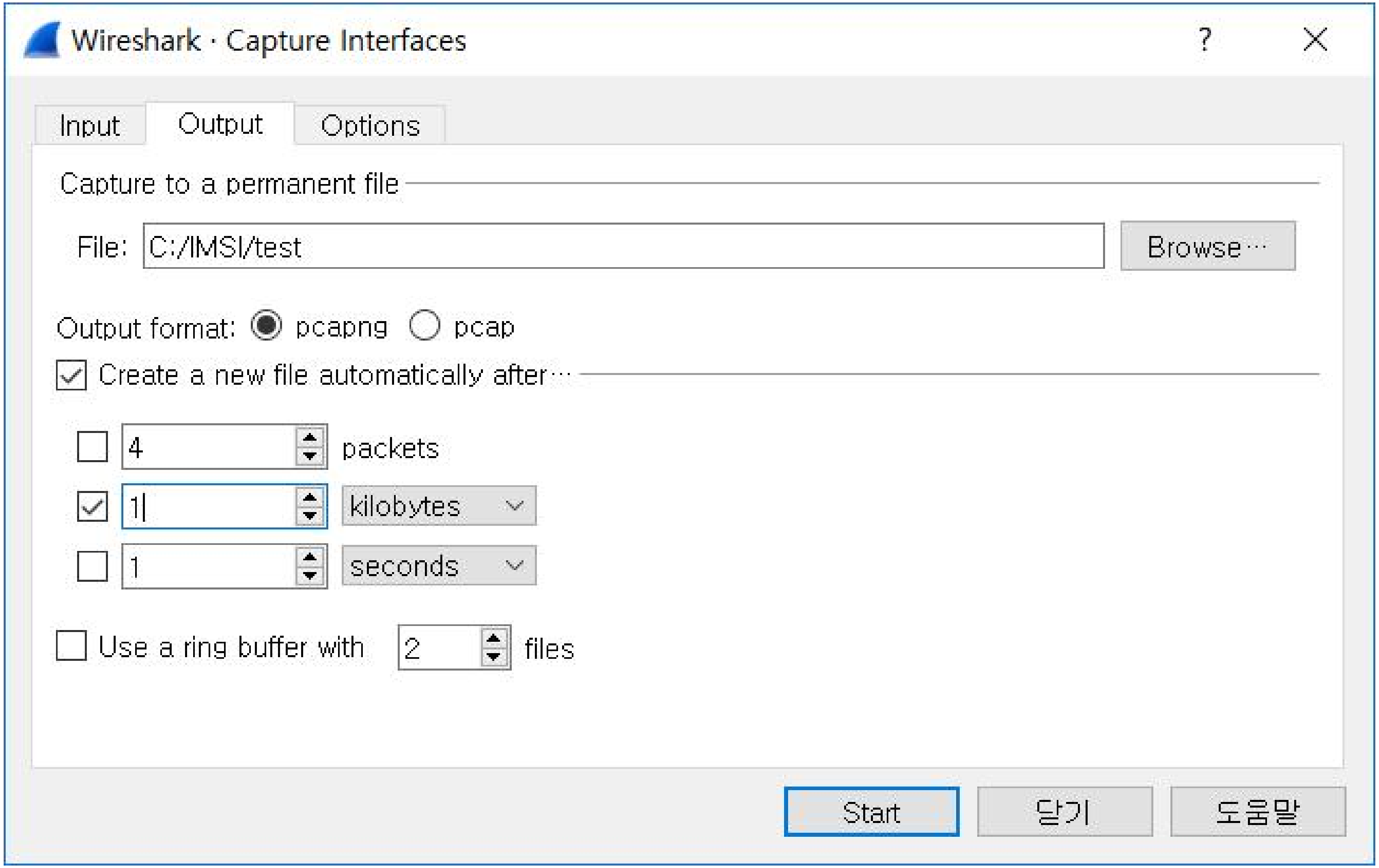
위의 어느 조건이든
먼저 만나면 파일 생성

파일 집합으로 수집

- Capture Options > Option Tab > Stop capture automatically after….



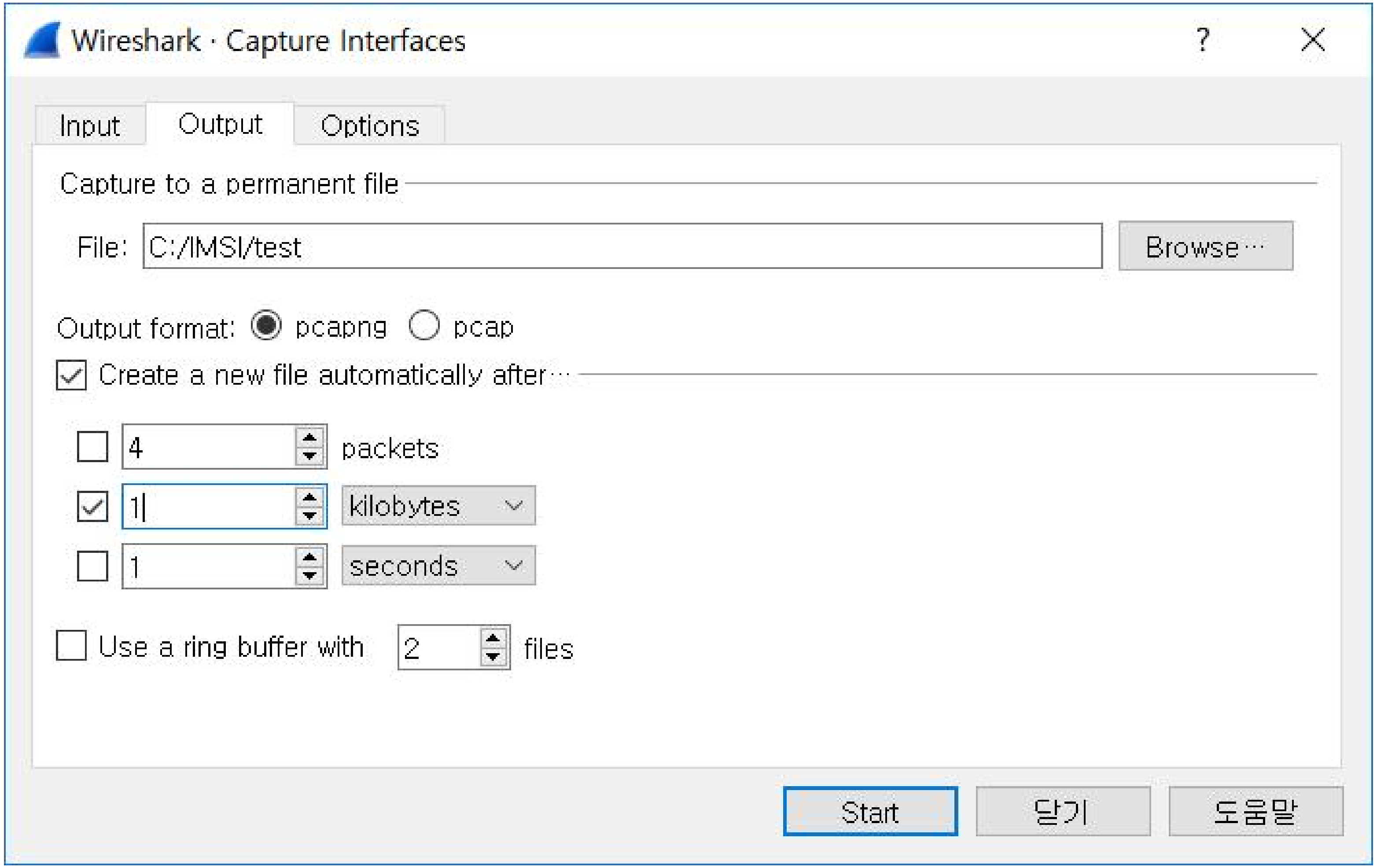
링 버퍼 사용



Wireshark · Capture Interfaces
Input
Output
Options
Capture to a permanent file
File:
C:/IMSI/test
Browse…
Output format:
pcapng
pcap
Create a new file automatically after…
4
packets
1
kilobytes
1
seconds
Use a ring buffer with
2
files
Start
닫기
도움말

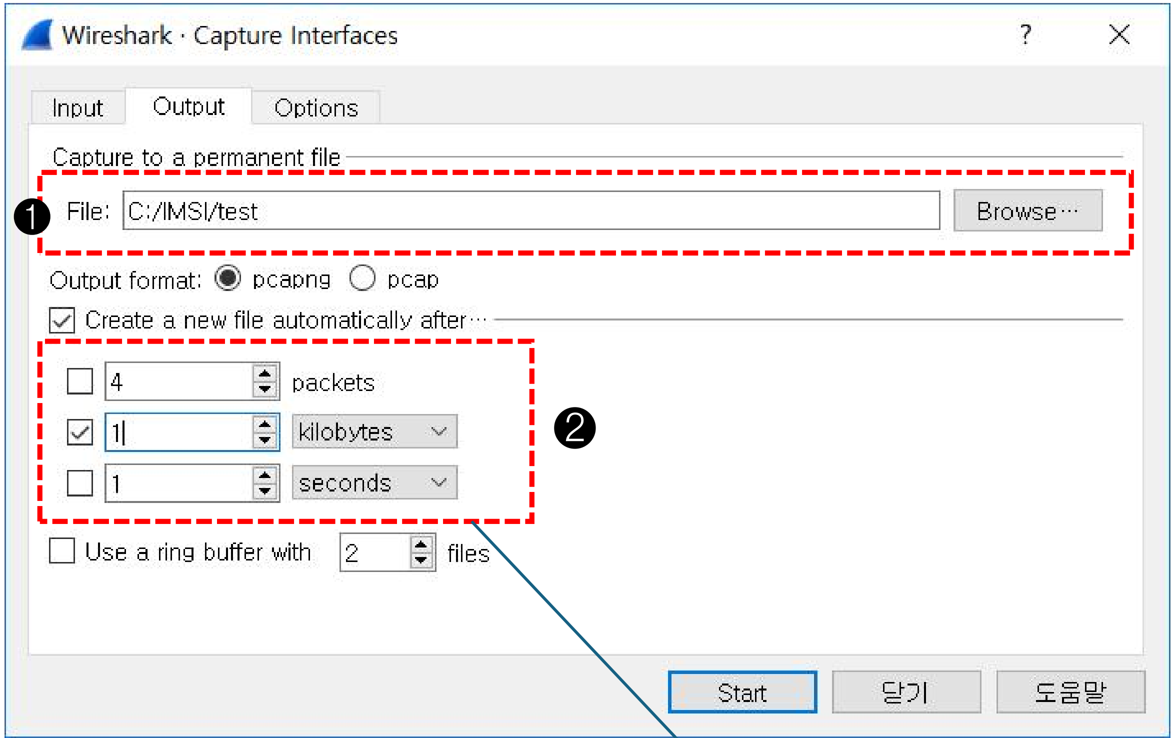
파일 집합으로 수집

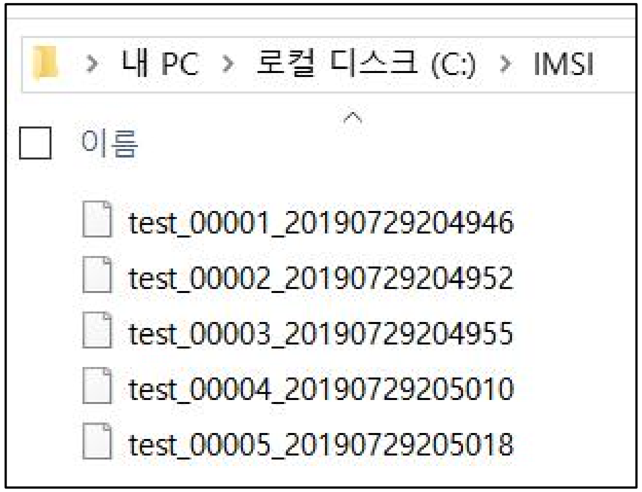
- Capture Options > Output Tab > Create a new file automatically after…



2. 파일 단위로 패킷 수집

- Capture Options > Output Tab > Create a new file automatically after…



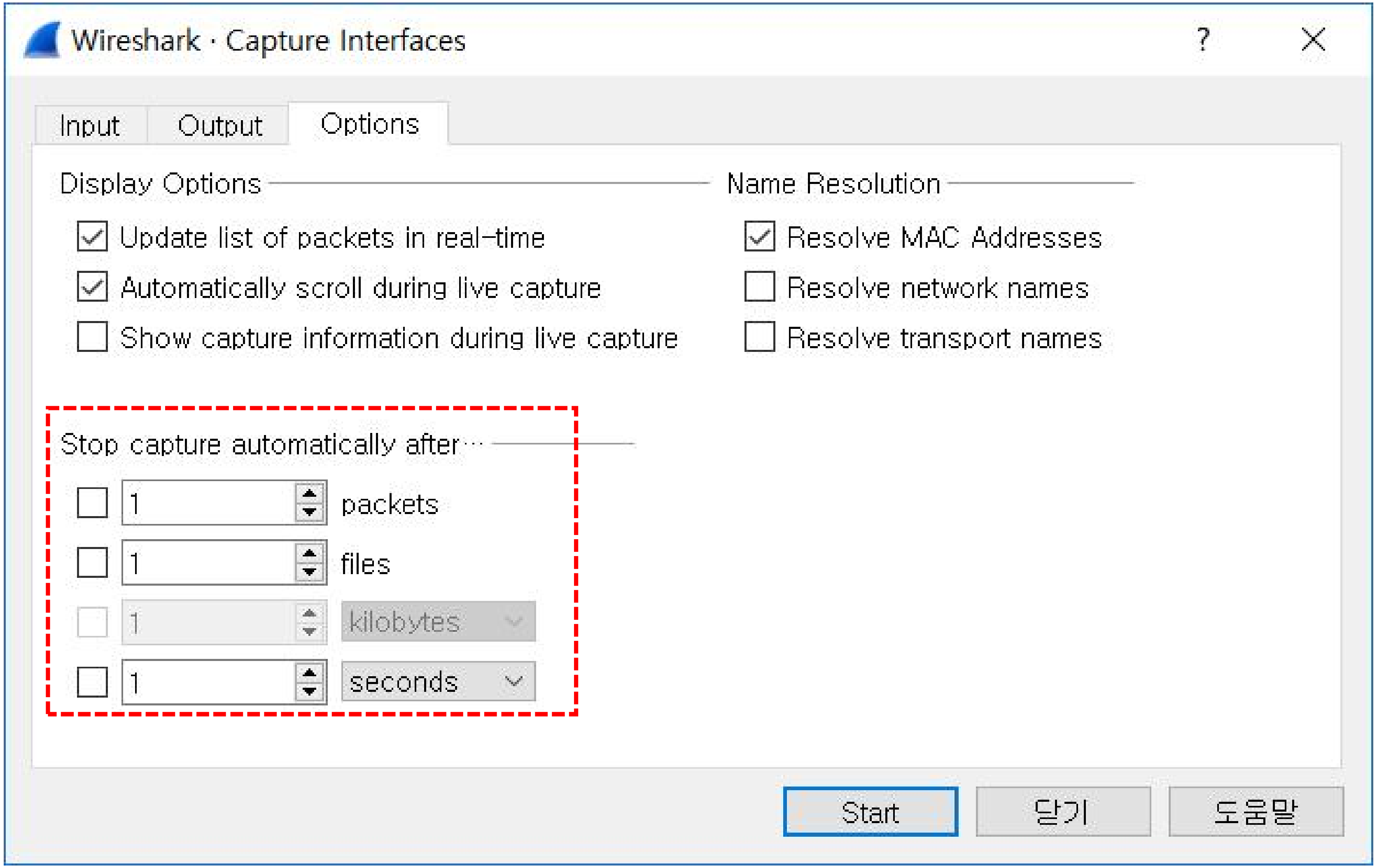


- File > Open
  - File > File set > List Files

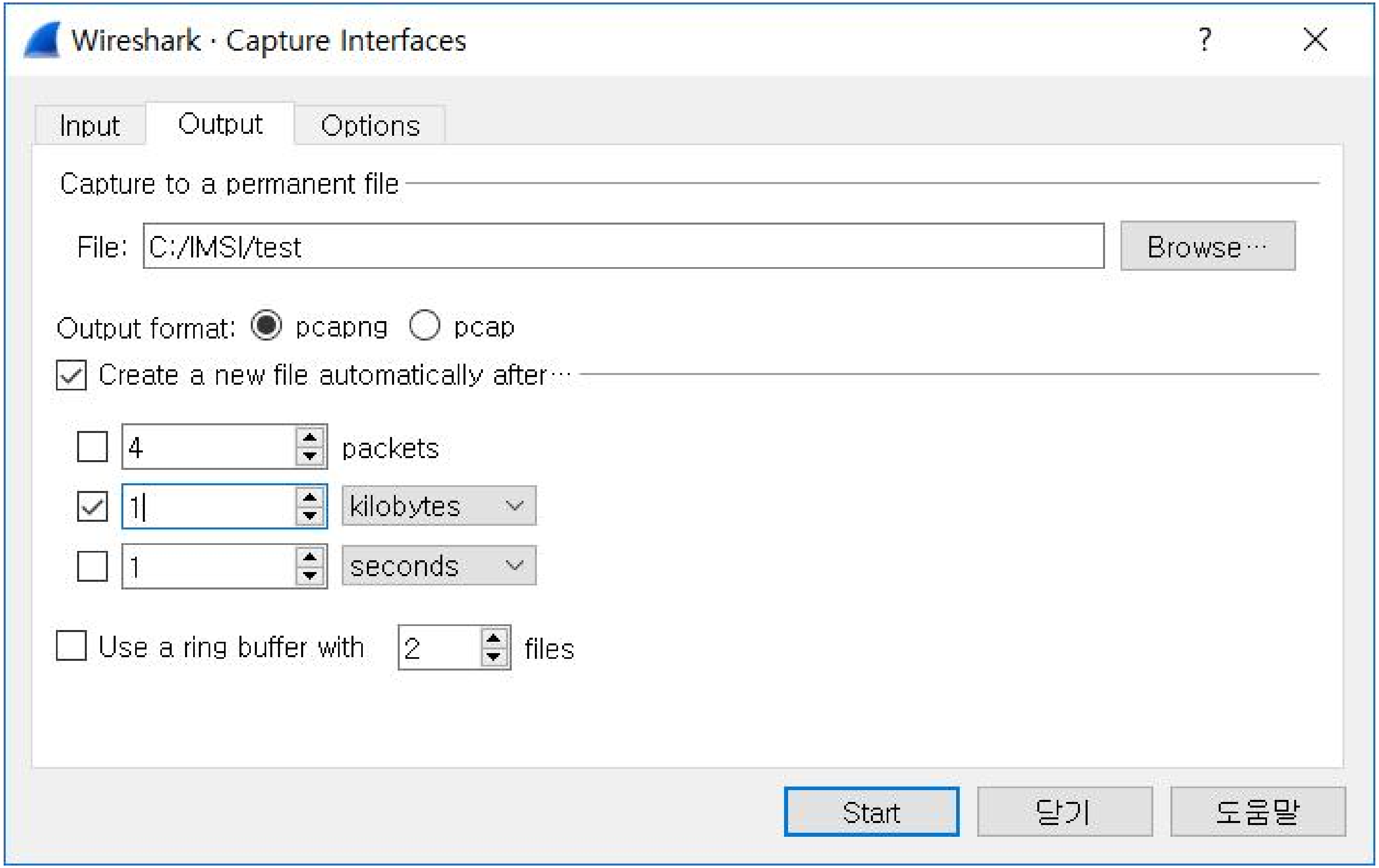
파일당 4개의 패킷
1MB 파일 크기
1초마다

위의 어느 조건이든
먼저 만나면 파일 생성

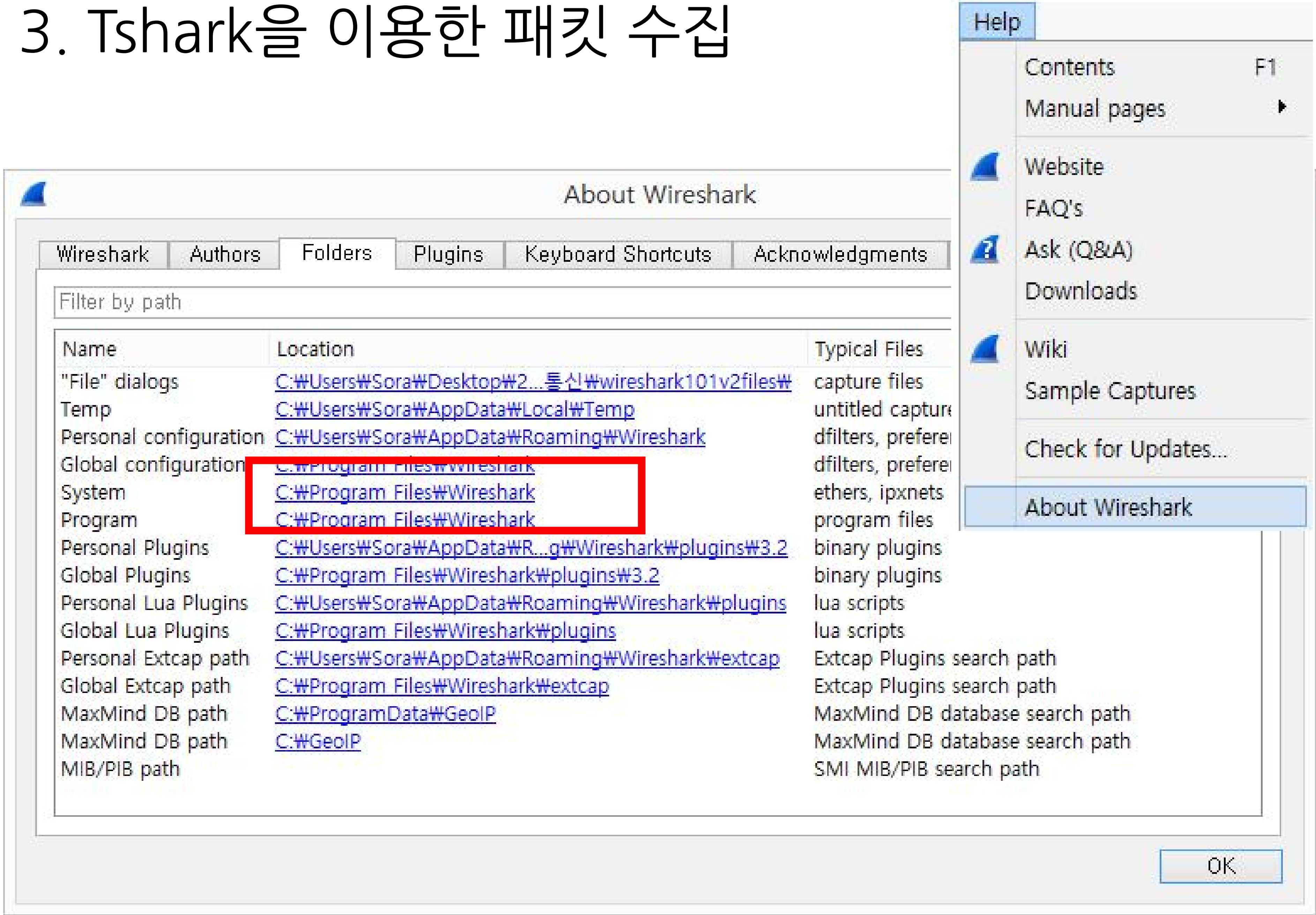
• Capture Options > Option Tab > Stop capture automatically after….

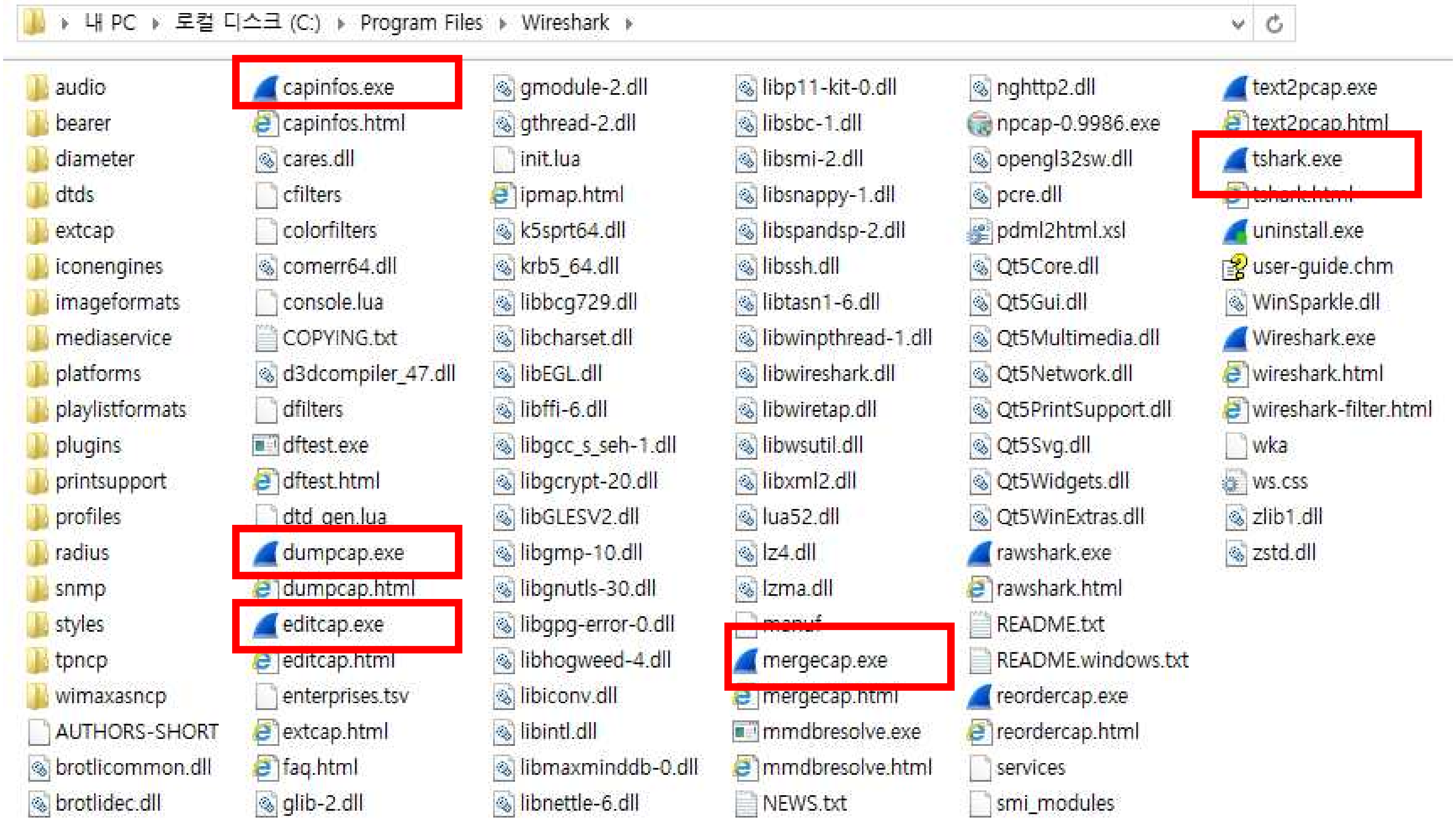


링 버퍼 사용


Wireshark · Capture Interfaces
?
Input
Output
Options
Capture to a permanent file
File:
C:/IMSI/test
Browse…
Output format:
pcapng
pcap
Create a new file automatically after…
4
packets
1
kilobytes
1
seconds
Use a ring buffer with
2
files
Start
닫기
도움말

3. Tshark을 이용한 패킷 수집




내 PC ▸ 로컬 디스크 (C:) ▸ Program Files ▸ Wireshark ▸
audio
bearer
diameter
dtds
extcap
iconengines
imageformats
mediaservice
platforms
playlistformats
plugins
printsupport
profiles
radius
snmp
styles
tpncp
wimaxasncp
AUTHORS-SHORT
brotlicommon.dll
brotlidec.dll
capinfos.exe
capinfos.html
cares.dll
cfilters
colorfilters
comerr64.dll
console.lua
COPYING.txt
d3dcompiler_47.dll
dfilters
dftest.exe
dftest.html
dtd_gen.lua
dumpcap.exe
dumpcap.html
editcap.exe
editcap.html
enterprises.tsv
extcap.html
faq.html
glib-2.dll
gmodule-2.dll
gthread-2.dll
init.lua
ipmap.html
k5sprt64.dll
krb5_64.dll
libbcg729.dll
libcharset.dll
libEGL.dll
libffi-6.dll
libgcc_s_seh-1.dll
libgcrypt-20.dll
libGLESV2.dll
libgmp-10.dll
libgnutls-30.dll
libgpg-error-0.dll
libhogweed-4.dll
libiconv.dll
libintl.dll
libmaxminddb-0.dll
libnettle-6.dll
libp11-kit-0.dll
libsbc-1.dll
libsmi-2.dll
libsnappy-1.dll
libspandsp-2.dll
libssh.dll
libtasn1-6.dll
libwinpthread-1.dll
libwireshark.dll
libwiretap.dll
libwsutil.dll
libxml2.dll
lua52.dll
lz4.dll
lzma.dll
manuf
mergecap.exe
mergecap.html
mmdbresolve.exe
mmdbresolve.html
NEWS.txt
nghttp2.dll
npcap-0.9986.exe
opengl32sw.dll
pcre.dll
pdml2html.xsl
Qt5Core.dll
Qt5Gui.dll
Qt5Multimedia.dll
Qt5Network.dll
Qt5PrintSupport.dll
Qt5Svg.dll
Qt5Widgets.dll
Qt5WinExtras.dll
rawshark.exe
rawshark.html
README.txt
README.windows.txt
reordercap.exe
reordercap.html
services
smi_modules
text2pcap.exe
text2pcap.html
tshark.exe
tshark.html
uninstall.exe
user-guide.chm
WinSparkle.dll
Wireshark.exe
wireshark.html
wireshark-filter.html
wka
ws.css
zlib1.dll
zstd.dll

커맨드라인에서 트래픽 수집

- dumpcap.exe나 tshark.exe를 이용해 커맨드 라인으로 트래픽 수집

  - Tshark를 구동하면 dumpcap.exe를 호출해서 수집 기능을 활용

```
C:\Program Files\Wireshark>dir dumpcap.exe

 C:\Program Files\Wireshark 디렉터리

2020-02-27  오전 05:47           420,416 dumpcap.exe
               1개 파일             420,416 바이트
               0개 디렉터리  86,740,180,992 바이트 남음

C:\Program Files\Wireshark>dir tshark.exe

 C:\Program Files\Wireshark 디렉터리

2020-02-27  오전 05:47           582,720 tshark.exe
               1개 파일             582,720 바이트
               0개 디렉터리  86,740,180,992 바이트 남음
```

❶ tshark -h

```
C:\Program Files\Wireshark>tshark -h
TShark (Wireshark) 3.0.3 (v3.0.3-0-g6130b92b0ec6)
Dump and analyze network traffic.
See https://www.wireshark.org for more information.

Usage: tshark [options] ...
```

❷ tshark -D

```
C:\Program Files\Wireshark>tshark -D
1.  \Device\NPF_{5256CF9B-1707-460C-B397-669CE852CFDE}
2.  \Device\NPF_{A378EED4-AC87-4168-8B22-DFDD0B9EC62E}
3.  \Device\NPF_{557124E1-28F6-4C2F-BDCE-1DFD75D011CE} (Wi-Fi)
4.  \Device\NPF_{630C09BF-1CAB-457F-978E-7A0BBEE76CF5}
5.  \Device\NPF_{E8E017EF-FC6D-4933-BD98-9AEC5CB26CF6}
6.  \Device\NPF_{27056470-1B11-4C61-8E92-0B25B7CB57C9}
7.  \Device\NPF_{DFBA2BCB-8989-40B4-873C-8CF862935F18}
8.  \Device\NPF_{6EC32DD5-41B3-45FF-B701-BB15DC1E7E45}
9.  \Device\NPF_{23E2556A-B636-483F-A13E-C3E889DD3825}
10.  \Device\NPF_{415D90F5-DFA5-426C-ABFE-BCCAB1542370}
```

❸ tshark -i 3

```
C:\Program Files\Wireshark>tshark -i 3
Capturing on 'Wi-Fi'
    1   0.000000 192.168.35.145 → 69.167.144.15 66 65160 64020 → https(443) [SYN] Seq=0 Win=65160 Len=0 MSS=1460 WS=256 SACK_PERM=1
    2   3.816065 192.168.35.145 → 121.53.202.218 66 65535 64021 → https(443) [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
    3   3.816070 192.168.35.145 → 121.53.202.218 66 65535 64022 → https(443) [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
    4   3.818776 121.53.202.218 → 192.168.35.145 66 8190 https(443) → 64022 [SYN, ACK] Seq=0 Ack=1 Win=8190 Len=0 MSS=1460 WS=16 SACK_PERM=1
    5   3.818895 192.168.35.145 → 121.53.202.218 54 1024 64022 → https(443) [ACK] Seq=1 Ack=1 Win=262144 Len=0
    6   3.819200 121.53.202.218 → 192.168.35.145 66 8190 https(443) → 64021 [SYN, ACK] Seq=0 Ack=1 Win=8190 Len=0 MSS=1460 WS=16 SACK_PERM=1
    7   3.819311 192.168.35.145 → 121.53.202.218 54 1024 64021 → https(443) [ACK] Seq=1 Ack=1 Win=262144 Len=0
    8   3.819713 192.168.35.145 → 121.53.202.218 295 1024 Client Hello
    9   3.819864 192.168.35.145 → 121.53.202.218 295 1024 Client Hello
```

* Ctrl+C 로 중단

❹ tshark -i 3 -w TST.pcapng

```
C:\Program Files\Wireshark>tshark -i 3 -w TST.pcapng
Capturing on 'Wi-Fi'
2472

C:\Program Files\Wireshark>
```

❺ tshark -r TST.pcapng

```
C:\Program Files\Wireshark>tshark -r TST.pcapng
    1   0.000000 192.168.35.145 → 69.167.144.15 74 65160 64159 → ht
    2   0.749769 192.168.35.145 → 104.74.232.184 54 256 64153 → htt
    3   0.752502 104.74.232.184 → 192.168.35.145 54 237 http(80) →
    4   0.752540 192.168.35.145 → 104.74.232.184 54 256 64153 → htt
    5   1.055385 192.168.35.145 → 110.76.141.124 66 64240 64160 → h
    6   1.058892 110.76.141.124 → 192.168.35.145 66 29200 https(443)
    7   1.058938 192.168.35.145 → 110.76.141.124 54 256 64160 → htt
    8   1.059047 192.168.35.145 → 110.76.141.124 256 256 Client Hell
```

❻ tshark -i 3 -a files:3 -b duration:10 -w myshark.pcapng

```
C:\Program Files\Wireshark>tshark -i 3 -a files:3 -b duration:10 -w myshark.pcapng
Capturing on 'Wi-Fi'
4464

C:\Program Files\Wireshark>dir myshark*
 C 드라이브의 볼륨에는 이름이 없습니다.
 볼륨 일련 번호: BCF8-FA59

 C:\Program Files\Wireshark 디렉터리

2018-08-12  오전 12:46           679,312 myshark_00001_20190801004608.pcapng
2018-08-12  오전 12:46         1,269,276 myshark_00002_20190801004618.pcapng
2018-08-12  오전 12:46           937,040 myshark_00003_20190801004629.pcapng
               3개 파일           2,885,628 바이트
               0개 디렉터리  126,970,032,128 바이트 남음
```

-a file:3 3개 파일 수집 후 자동 정지

-b duration:10 10초후에 다음 파일을 생성

-w myshark.pcapng 추적파일명

커맨드라인 수집 과정에서 수집 필터(캡처필터)

❶ tshark -i 3 -f “tcp port 443” -w mysecport443.pcapng

```
C:\Program Files\Wireshark>tshark -i 3 -f "tcp port 443" -w mysecport443.pcapng
Capturing on 'Wi-Fi'
2734

C:\Program Files\Wireshark>tshark -r mysecport443.pcapng
    1   0.000000 192.168.35.145 → 69.167.144.15 74 65160 64805 → https(443) [SYN] Seq=0 Wi
    2   3.010265 192.168.35.145 → 69.167.144.15 74 65160 [TCP Retransmission] 64805 → htt
    3   3.627403 192.168.35.145 → 52.175.23.79 66 64240 64806 → https(443) [SYN] Seq=0 Wi
    4   3.710985 52.175.23.79 → 192.168.35.145 66 8192 https(443) → 64806 [SYN, ACK] Seq=
    5   3.711076 192.168.35.145 → 52.175.23.79 54 258 64806 → https(443) [ACK] Seq=1 Ack=
    6   3.711563 192.168.35.145 → 52.175.23.79 274 258 Client Hello
    7   3.792328 52.175.23.79 → 192.168.35.145 1514 1026 [TCP segment of a reassembled PDU]
    8   3.792330 52.175.23.79 → 192.168.35.145 1514 1026 https(443) → 64806 [ACK] Seq=146
    9   3.792335 52.175.23.79 → 192.168.35.145 538 1026 Server Hello, Certificate, Server
```

❷ tshark -i 3 -f “tcp port 443 and host 192.168.1.1” -w my443.pcapng

커맨드라인 수집 과정에서 디스플레이 필터

❶ tshark -r “mysecport443.pcapng” -Y “tcp.analysis.flags”

```
C:\Program Files\Wireshark>tshark -r "mysecport443.pcapng" -Y "tcp.analysis.flags"
    2   3.010265 192.168.35.145 → 69.167.144.15 74 65160 [TCP Retransmission] 64805 → https(443)
 1113   9.019700 192.168.35.145 → 69.167.144.15 66 65160 [TCP Retransmission] 64805 → https(443)
 2734  24.038230 192.168.35.145 → 69.167.144.15 74 65160 [TCP Retransmission] 64898 → https(443)
```

❷ tshark -r “mysecport443.pcapng” -Y “tcp.analysis.flags” -w tcpflag.pcapng

❸ tshark -r “mysecport443.pcapng” -Y “http.request.method == GET”

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http://github.com/soraddang